Custom Neural Networks

L-layers Vision Transformer Encoder

Anomaly Detection in Industrial Images

https://github.com/lucacareddu/Custom-Neural-Networks

1 Forward

$\boldsymbol{X} \in \mathbb{R}^{n \times m}$, $\texttt{keep} \sim \text{Bern}\,(p)^n$, $\boldsymbol{W}_{in} \in \mathbb{R}^{m \times d}$, $\texttt{pos} \in \mathbb{R}^{n \times d}$, $\epsilon \in \mathbb{R}$, $\boldsymbol{\gamma}^{(l)}, \boldsymbol{\beta}^{(l)} \in \mathbb{R}^d$ with $l = 1, \dots, L+1$, $\boldsymbol{W}^{(l)}_{\{Q,K,V,O\}} \in \mathbb{R}^{d \times d}$ with $l = 1, \dots, L$, $\boldsymbol{W}_{out} \in \mathbb{R}^{d \times m}$

$$\tilde{\boldsymbol{X}} = \boldsymbol{X} \odot \texttt{keep} \qquad \in \mathbb{R}^{n \times m} \tag{1.1}$$

$$\boldsymbol{h}^{(0)} = \tilde{\boldsymbol{X}} \boldsymbol{W}_{in} + \texttt{pos} \qquad \in \mathbb{R}^{n \times d} \tag{1.2}$$

Then, for $l = 1, \dots, L$,

$$\boldsymbol{\sigma}^{(l)} = \sqrt{\mathbb{E}_d\big((\boldsymbol{h}^{(l-1)} - \mathbb{E}_d(\boldsymbol{h}^{(l-1)}))^{\circ 2}\big) + \epsilon} \qquad \in \mathbb{R}^{n} \tag{1.3}$$

$$\hat{\boldsymbol{h}}^{(l)} = \frac{\boldsymbol{h}^{(l-1)} - \mathbb{E}_d(\boldsymbol{h}^{(l-1)})}{\boldsymbol{\sigma}^{(l)}} \qquad \in \mathbb{R}^{n \times d} \tag{1.4}$$

$$\boldsymbol{a}^{(l)} = \hat{\boldsymbol{h}}^{(l)} \odot \boldsymbol{\gamma}^{(l)} + \boldsymbol{\beta}^{(l)} \qquad \in \mathbb{R}^{n \times d} \tag{1.5}$$

$$\boldsymbol{M}^{(l)} = \boldsymbol{a}^{(l)} \boldsymbol{W}^{(l)}_{M}, \quad M \in \{Q, K, V\} \qquad \in \mathbb{R}^{n \times d} \tag{1.6}$$

$$\boldsymbol{S}^{(l)} = \text{Softmax}\left(\frac{\boldsymbol{Q}^{(l)} \boldsymbol{K}^{(l)\intercal}}{\sqrt{d}}\right) \qquad \in \mathbb{R}^{n \times n} \tag{1.7}$$

$$\boldsymbol{U}^{(l)} = \boldsymbol{S}^{(l)} \boldsymbol{V}^{(l)} \qquad \in \mathbb{R}^{n \times d} \tag{1.8}$$

$$\boldsymbol{h}^{(l)} = \boldsymbol{U}^{(l)} + \boldsymbol{a}^{(l)} \boldsymbol{W}^{(l)}_{O} + \boldsymbol{h}^{(l-1)} \qquad \in \mathbb{R}^{n \times d} \tag{1.9}$$

and finally

$$\boldsymbol{\sigma}^{(L+1)} = \sqrt{\mathbb{E}_d\big((\boldsymbol{h}^{(L)} - \mathbb{E}_d(\boldsymbol{h}^{(L)}))^{\circ 2}\big) + \epsilon} \qquad \in \mathbb{R}^{n} \tag{1.10}$$

$$\hat{\boldsymbol{h}}^{(L+1)} = \frac{\boldsymbol{h}^{(L)} - \mathbb{E}_d(\boldsymbol{h}^{(L)})}{\boldsymbol{\sigma}^{(L+1)}} \qquad \in \mathbb{R}^{n \times d} \tag{1.11}$$

$$\boldsymbol{a}^{(L+1)} = \hat{\boldsymbol{h}}^{(L+1)} \odot \boldsymbol{\gamma}^{(L+1)} + \boldsymbol{\beta}^{(L+1)} \qquad \in \mathbb{R}^{n \times d} \tag{1.12}$$

$$\hat{\boldsymbol{X}} = \boldsymbol{a}^{(L+1)} \boldsymbol{W}_{out} \qquad \in \mathbb{R}^{n \times m} \tag{1.13}$$

$$\mathcal{L} = \frac{1}{2} \mathbb{E}_{n,m}\left((\hat{\boldsymbol{X}} - \boldsymbol{X})^{\circ 2}\right) \qquad \in \mathbb{R} \tag{1.14}$$

function $\mathcal{L}(\boldsymbol{X}, \texttt{keep}, \boldsymbol{W}_{in}, \texttt{pos}, \epsilon, \boldsymbol{\gamma}^{(1:L+1)}, \boldsymbol{\beta}^{(1:L+1)}, \boldsymbol{W}^{(1:L)}_{\{Q,K,V,O\}}, \boldsymbol{W}_{out})$
  $\tilde{\boldsymbol{X}} \leftarrow \boldsymbol{X} \odot \texttt{keep}$
  $\boldsymbol{h}^{(0)} \leftarrow \tilde{\boldsymbol{X}}\boldsymbol{W}_{in} + \texttt{pos}$
  $\boldsymbol{h}^{(L)} \leftarrow \text{SelfAttention}^L(\boldsymbol{h}^{(0)}, \boldsymbol{\gamma}^{(1:L)}, \boldsymbol{\beta}^{(1:L)}, \epsilon, \boldsymbol{W}^{(1:L)}_{\{Q,K,V,O\}})$
  $\hat{\boldsymbol{X}} \leftarrow \text{LayerNorm}(\boldsymbol{h}^{(L)}, \boldsymbol{\gamma}^{(L+1)}, \boldsymbol{\beta}^{(L+1)}, \epsilon)\boldsymbol{W}_{out}$
  $\mathcal{L} \leftarrow \text{MSE}(\boldsymbol{X}, \hat{\boldsymbol{X}})$
  return $\mathcal{L}$
end function

2 Backward

$$\nabla_{\hat{X}} \overset{(1.14)}{=} \frac{1}{nm}(\hat{\boldsymbol{X}} - \boldsymbol{X}) \tag{2.1}$$

$$\nabla_{W_{out}} \overset{(1.13)}{=} \boldsymbol{a}^{(L+1)\intercal}\nabla_{\hat{X}} \tag{2.2}$$

$$\nabla_{a^{(L+1)}} \overset{(1.13)}{=} \nabla_{\hat{X}}\boldsymbol{W}_{out}^{\intercal} \tag{2.3}$$

$$\nabla_{\beta^{(L+1)}} \overset{(1.12)}{=} \nabla_{a^{(L+1)}}{}^{\intercal}\mathbf{1}_n \tag{2.4}$$

$$\nabla_{\gamma^{(L+1)}} \overset{(1.12)}{=} (\nabla_{a^{(L+1)}} \odot \hat{\boldsymbol{h}}^{(L+1)})^{\intercal}\mathbf{1}_n \tag{2.5}$$

$$\nabla_{\hat{h}^{(L+1)}} \overset{(1.12)}{=} \nabla_{a^{(L+1)}} \odot \mathbf{1}_n\boldsymbol{\gamma}^{(L+1)\intercal} \tag{2.6}$$

With the broadcast along d kept implicit for ease of reading,

$$\nabla_{h^{(L)}} \overset{(1.11)}{=} \frac{1}{\boldsymbol{\sigma}^{(L+1)}} \odot \left(\nabla_{\hat{h}^{(L+1)}} - \mathbb{E}_d(\nabla_{\hat{h}^{(L+1)}}) - \hat{\boldsymbol{h}}^{(L+1)} \odot \mathbb{E}_d(\nabla_{\hat{h}^{(L+1)}} \odot \hat{\boldsymbol{h}}^{(L+1)})\right) \tag{2.7}$$

The row-wise derivation of (2.7) is deferred to Appendix A.

Then, for $l = L, \ldots, 1$,

$$\nabla_{W_O^{(l)}} \overset{(1.9)}{=} \boldsymbol{a}^{(l)\intercal}\nabla_{h^{(l)}} \tag{2.8}$$

$$\nabla_{V^{(l)}} \overset{(1.8)}{=} \boldsymbol{S}^{(l)\intercal}\nabla_{h^{(l)}} \tag{2.9}$$

$$\nabla_{S^{(l)}} \overset{(1.8)}{=} \nabla_{h^{(l)}}\boldsymbol{V}^{(l)\intercal} \tag{2.10}$$

Now, with $\boldsymbol{s}_i^{(l)}$ the i-th row of $\boldsymbol{S}^{(l)}$ as a column vector,

$$\frac{\partial \boldsymbol{s}_i^{(l)}}{\partial(\boldsymbol{Q}^{(l)}\boldsymbol{K}^{(l)\intercal})_i} \overset{(1.7)}{=} \frac{1}{\sqrt{d}}\left(\text{diag}(\boldsymbol{s}_i^{(l)}) - \boldsymbol{s}_i^{(l)}\boldsymbol{s}_i^{(l)\intercal}\right) \tag{2.11}$$

contracted row-wise and stacked, it follows that

$$\nabla_{Q^{(l)}K^{(l)\intercal}} \overset{(1.7)}{=} \frac{1}{\sqrt{d}}\boldsymbol{S}^{(l)} \odot \left(\nabla_{S^{(l)}} - ((\nabla_{S^{(l)}} \odot \boldsymbol{S}^{(l)})\mathbf{1}_n)\mathbf{1}_n^{\intercal}\right) \tag{2.12}$$

$$\nabla_{K^{(l)}} \overset{(1.7)}{=} (\nabla_{Q^{(l)}K^{(l)\intercal}})^{\intercal}\boldsymbol{Q}^{(l)} \tag{2.13}$$

$$\nabla_{Q^{(l)}} \overset{(1.7)}{=} \nabla_{Q^{(l)}K^{(l)\intercal}}\boldsymbol{K}^{(l)} \tag{2.14}$$

Then we have

$$\nabla_{W_M^{(l)}} \overset{(1.6)}{=} \boldsymbol{a}^{(l)\intercal} \nabla_{M^{(l)}}, \quad M \in \{Q, K, V\} \tag{2.15}$$

$$\nabla_{a^{(l)}} \overset{(1.6),(1.9)}{=} \nabla_{Q^{(l)}} \boldsymbol{W}_Q^{(l)\intercal} + \nabla_{K^{(l)}} \boldsymbol{W}_K^{(l)\intercal} + \nabla_{V^{(l)}} \boldsymbol{W}_V^{(l)\intercal} + \nabla_{h^{(l)}} \boldsymbol{W}_O^{(l)\intercal} \tag{2.16}$$

$$\nabla_{\beta^{(l)}} \overset{(1.5)}{=} \nabla_{a^{(l)}}{}^{\intercal} \mathbf{1}_n \tag{2.17}$$

$$\nabla_{\gamma^{(l)}} \overset{(1.5)}{=} (\nabla_{a^{(l)}} \odot \hat{\boldsymbol{h}}^{(l)})^{\intercal} \mathbf{1}_n \tag{2.18}$$

$$\nabla_{\hat{h}^{(l)}} \overset{(1.5)}{=} \nabla_{a^{(l)}} \odot \mathbf{1}_n \boldsymbol{\gamma}^{(l)\intercal} \tag{2.19}$$

and, with the residual branch contributing the identity term $\nabla_{h^{(l)}}$ and the same implicit broadcast,

$$\nabla_{h^{(l-1)}} \overset{(1.4),(1.9)}{=} \nabla_{h^{(l)}} + \frac{1}{\boldsymbol{\sigma}^{(l)}} \odot \left(\nabla_{\hat{h}^{(l)}} - \mathbb{E}_d(\nabla_{\hat{h}^{(l)}}) - \hat{\boldsymbol{h}}^{(l)} \odot \mathbb{E}_d(\nabla_{\hat{h}^{(l)}} \odot \hat{\boldsymbol{h}}^{(l)}) \right) \tag{2.20}$$

The row-wise derivation of (2.20) is deferred to Appendix A.

Finally,

$$\nabla_{\texttt{pos}} \overset{(1.2)}{=} \nabla_{h^{(0)}} \tag{2.21}$$

$$\nabla_{W_{in}} \overset{(1.2)}{=} \tilde{\boldsymbol{X}}^{\intercal} \nabla_{h^{(0)}} \tag{2.22}$$

Appendix

A LayerNorm backward

Equations (2.7) and (2.20) follow by differentiating the row-wise normalization (1.3)–(1.4) one row at a time and stacking, exactly as (2.12) follows from (2.11).
Now, with $\boldsymbol{h}_i^{(l-1)}$, $\hat{\boldsymbol{h}}_i^{(l)}$ and $\nabla_{\hat{h}_i^{(l)}}$ the i-th rows of $\boldsymbol{h}^{(l-1)}$, $\hat{\boldsymbol{h}}^{(l)}$ and $\nabla_{\hat{h}^{(l)}}$ as column vectors, and $\sigma_i^{(l)}$ the i-th entry of $\boldsymbol{\sigma}^{(l)}$, equations (1.3) and (1.4) read

$$\sigma_i^{(l)} \overset{(1.3)}{=} \sqrt{\frac{1}{d}\left\|\left(\boldsymbol{I}_d - \frac{1}{d}\mathbf{1}_d\mathbf{1}_d^\intercal\right)\boldsymbol{h}_i^{(l-1)}\right\|^2 + \epsilon} \tag{A.1}$$

$$\hat{\boldsymbol{h}}_i^{(l)} \overset{(1.4)}{=} \frac{1}{\sigma_i^{(l)}}\left(\boldsymbol{I}_d - \frac{1}{d}\mathbf{1}_d\mathbf{1}_d^\intercal\right)\boldsymbol{h}_i^{(l-1)} \tag{A.2}$$

Differentiating (A.1), with the centering matrix $\boldsymbol{I}_d - \frac{1}{d}\mathbf{1}_d\mathbf{1}_d^\intercal$ symmetric and idempotent, and ϵ constant gives

$$\frac{\partial \sigma_i^{(l)}}{\partial \boldsymbol{h}_i^{(l-1)}} \overset{(A.1)}{=} \frac{1}{d\,\sigma_i^{(l)}}\left(\boldsymbol{I}_d - \frac{1}{d}\mathbf{1}_d\mathbf{1}_d^\intercal\right)\boldsymbol{h}_i^{(l-1)} \overset{(A.2)}{=} \frac{1}{d}\hat{\boldsymbol{h}}_i^{(l)} \tag{A.3}$$

so that differentiating (A.2) by the quotient rule, with (A.3) supplying the derivative of $\sigma_i^{(l)}$ and (A.2) again supplying $\left(\boldsymbol{I}_d - \frac{1}{d}\mathbf{1}_d\mathbf{1}_d^\intercal\right)\boldsymbol{h}_i^{(l-1)} = \sigma_i^{(l)}\hat{\boldsymbol{h}}_i^{(l)}$,

$$\frac{\partial \hat{\boldsymbol{h}}_i^{(l)}}{\partial \boldsymbol{h}_i^{(l-1)}} \overset{(A.2)}{=} \frac{1}{\sigma_i^{(l)}}\left(\boldsymbol{I}_d - \frac{1}{d}\mathbf{1}_d\mathbf{1}_d^\intercal - \frac{1}{d}\hat{\boldsymbol{h}}_i^{(l)}\hat{\boldsymbol{h}}_i^{(l)\intercal}\right) \tag{A.4}$$

The Jacobian is symmetric, so contracting it with $\nabla_{\hat{h}_i^{(l)}}$ needs no transposition

$$\nabla_{h_i^{(l-1)}} \overset{(A.4)}{=} \frac{1}{\sigma_i^{(l)}}\left(\nabla_{\hat{h}_i^{(l)}} - \frac{1}{d}\mathbf{1}_d\mathbf{1}_d^\intercal\nabla_{\hat{h}_i^{(l)}} - \frac{1}{d}\hat{\boldsymbol{h}}_i^{(l)}\hat{\boldsymbol{h}}_i^{(l)\intercal}\nabla_{\hat{h}_i^{(l)}}\right) \tag{A.5}$$

and, stacked over $i = 1, \ldots, n$, the two inner products become the row-wise means $\mathbb{E}_d(\nabla_{\hat{h}^{(l)}})$ and $\mathbb{E}_d(\nabla_{\hat{h}^{(l)}} \odot \hat{\boldsymbol{h}}^{(l)})$ broadcast along d, giving back (2.7) for $l = L+1$ and (2.20) for $l = 1, \ldots, L$, there summed with the identity term $\nabla_{h^{(l)}}$ of the residual branch.